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Instructions Follow instructions *carefully*; failure to do so will result in points being deducted.

1. 8 points Prove, using induction, that

$$1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$$

whenever n is a positive integer.

- State and prove the basis step.
- State the inductive hypothesis.
- State the inductive conclusion.
- Prove the inductive conclusion by the method of induction. You must justify the relevant steps.

Solution: Enter your solution here:

A: Base case $n = 1$:

$$1^4 = \frac{1(1+1)(2 \cdot 1 + 1)(3 \cdot 1^2 + 3 \cdot 1 - 1)}{30}$$

$$\sum_{n=1}^1 n^4 = 1 = 1\checkmark \quad \frac{1 \cdot 2 \cdot 3 \cdot 5}{30} = \frac{30}{30} = 1\checkmark$$

B: Inductive hypothesis

$$1^4 + 2^4 + 3^4 + \cdots + k^4 = \frac{k(k+1)(2k+1)(3k^2+3k-1)}{30}$$

C: Inductive conclusion:

$$\underbrace{1^4 + 2^4 + 3^4 + \cdots + k^4}_{\text{IH}} + (k+1)^4 = \frac{(k+1)(k+2)(2k+3)(3(k+1)^2 + 3(k+1) - 1)}{30}$$

D: Prove:

$$\frac{k(k+1)(2k+1)(3k^2+3k-1)}{30} + (k+1)^4 = \frac{(k+1)(k+2)(2k+3)(3(k+1)^2 + 3(k+1) - 1)}{30}$$

$$\frac{k(k+1)(2k+1)(3k^2+3k-1) + 30(k+1)^4}{30}$$

$$\frac{k+1}{30} [k(2k+1)(3k^2+3k-1) + 30(k+1)^3]$$

Expanding $(2k^2+k)(3k^2+3k-1)$:

$$6k^4 + 6k^3 - 2k^2 + 3k^3 + 3k^2 - k = 6k^4 + 9k^3 + k^2 - k$$

And $30(k+1)^3 = 30k^3 + 90k^2 + 90k + 30$

Combining:

$$\frac{k+1}{30} [6k^4 + 39k^3 + 91k^2 + 89k + 30]$$

We now expand the right-hand side target $(k+2)(2k+3)(3k^2+9k+5)$ to verify it equals the polynomial above.

First, $(k+2)(2k+3) = 2k^2 + 7k + 6$.

Then $(2k^2 + 7k + 6)(3k^2 + 9k + 5)$:

$$\begin{aligned} &= 6k^4 + 18k^3 + 10k^2 + 21k^3 + 63k^2 + 35k + 18k^2 + 54k + 30 \\ &= 6k^4 + 39k^3 + 91k^2 + 89k + 30 \checkmark \end{aligned}$$

Since $3(k+1)^2 + 3(k+1) - 1 = 3k^2 + 9k + 5$, we conclude:

$$\frac{k+1}{30} (k+2)(2k+3)(3k^2+9k+5) = \frac{(k+1)(k+2)(2k+3)(3(k+1)^2 + 3(k+1) - 1)}{30}$$

2. 8 points Prove the following using induction whenever n is a positive integer.

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1}.$$

- (a) State and prove the basis step.
- (b) State the inductive hypothesis.
- (c) State the inductive conclusion.
- (d) Prove the inductive conclusion by the method of induction. You must justify the relevant steps.

Solution: Enter your solution here:

A: Base case $n = 1$:

$$\frac{1}{1(1+1)} = 1 - \frac{1}{1+1} \quad \frac{1}{2} = \frac{1}{2} \checkmark$$

B: Inductive hypothesis:

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{k(k+1)} = 1 - \frac{1}{k+1}$$

C: Inductive conclusion:

$$\underbrace{\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{k(k+1)}}_{\text{IH}} + \frac{1}{(k+1)(k+2)} = 1 - \frac{1}{k+2}$$

D: Prove:

$$1 - \frac{1}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$\begin{aligned}
&= 1 + \frac{1}{k+1} \left(-1 + \frac{1}{k+2} \right) \\
&= 1 + \frac{1}{k+1} \left(\frac{-(k+2) + 1}{k+2} \right) \\
&= 1 + \frac{1}{k+1} \left(\frac{-(k+1)}{k+2} \right) \\
&= 1 - \frac{1}{k+2}
\end{aligned}$$

3. 8 points Prove, using induction, that 2 divides $n^3 + 3n$ whenever n is a positive integer.

- (a) State and prove the basis step.
- (b) State the inductive hypothesis.
- (c) State the inductive conclusion.
- (d) Prove the inductive conclusion by the method of induction. You must justify the relevant steps.

Solution: Enter your solution here:

A: Show $2 \mid n^3 + 3n$.

Base case $n = 1$: $2 \mid 1^3 + 3(1) = 4 = 2 \cdot 2 \checkmark$

B: Inductive hypothesis: Assume $2 \mid k^3 + 3k$.

C: Show: $2 \mid (k+1)^3 + 3(k+1)$.

D: Expanding:

$$(k+1)^3 + 3(k+1) = k^3 + 3k^2 + 3k + 1 + 3k + 3$$

$$= (3k^2 + 6k + 4) + \underbrace{k^3 + 3k}_{2M}$$

$$= 2 \underbrace{\left(\frac{3}{2}k^2 + 3k + 2 \right)}_a + 2M$$

$$= 2a + 2M = 2(a + M) = 2C$$

4. 8 points An employee joined a company in 2020 with a starting salary of \$95,000. Every year, this employee receives a raise of \$1000 plus 3% of the salary of the previous year.

- (a) Set up a recurrence relation for this employee's salary n years after 2020. **Justify** your answer.
- (b) Using the iterative method, solve the recurrence relation for this employee's salary n years after 2020.
- (c) Determine the salary of this employee in 2026.

Solution: Enter your solution here:

$$P_0 = 95000$$

$$P_1 = 95000(1.03) + 1000$$

$$P_2 = [95000(1.03) + 1000](1.03) + 1000$$

$$P_3 = P_2(1.03) + 1000$$

$$P_n = P_{n-1}(1.03) + 1000$$

$$P_n = 1.03^n(95000) + 1000 \left[\frac{1.03^n - 1}{0.03} \right]$$

$$P_6 = 1.03^6(95000) + 1000 \left[\frac{1.03^6 - 1}{0.03} \right] \approx 119,903$$

5. 12 points For the following recurrences, provide an asymptotic characterization using the **Master theorem**.

Note: Clearly show the values of the constants of the recurrence relation when using the Master theorem, and determine which case of the Master theorem applies to this recurrence relation.

(a) $T(n) = 2T(n/3) + 4$

(b) $T(n) = 3T(n/3) + 3n$

(c) $T(n) = 2T(n/2) + 3n^2$

(d) $T(n) = 8T(n/2) + n^3$

Solution: Enter your solution here:

A: $a = 2, \quad b = 3, \quad d = 0$

$$2 > 3^0 = 1$$

$$T(n) \in \mathcal{O}(n^{\log_3 2})$$

B: $a = 3, \quad b = 3, \quad d = 1$

$$3 = 3^1$$

$$T(n) \in \mathcal{O}(n \log n)$$

C: $a = 2, \quad b = 2, \quad d = 2$

$$2 < 2^2 = 4$$

$$T(n) \in \mathcal{O}(n^2)$$

D: $a = 8, \quad b = 2, \quad d = 3$

$$8 = 2^3$$

$$T(n) \in \theta(n^3 \log n)$$